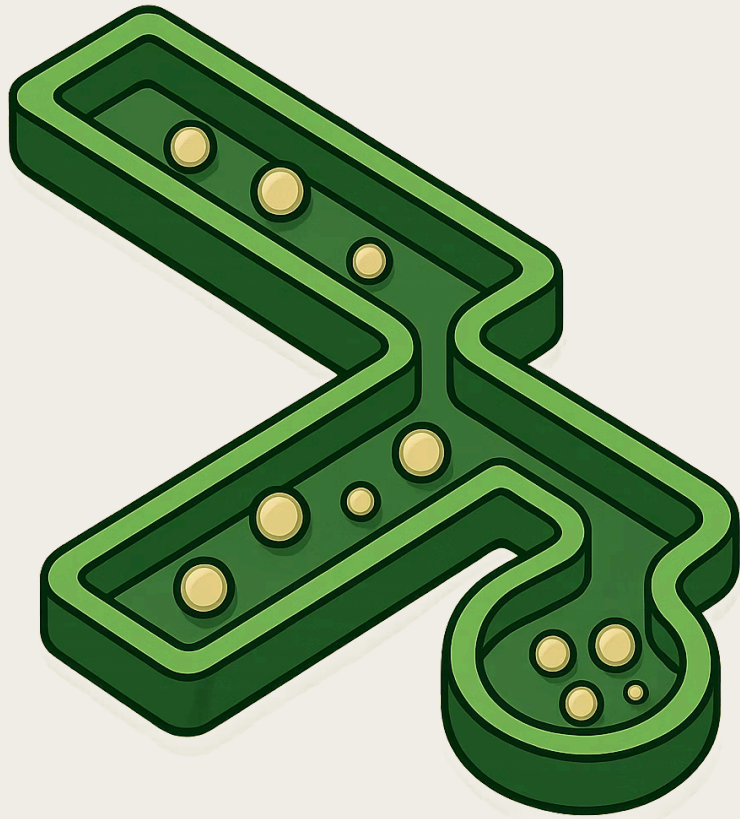




Overcoming Inter-Symbol Interference in Molecular Communication

From Mathematical Cancellation to Machine Learning

ALI AYHAN GÜNDER & SAMED KIZILHAN

Paper 1

Meng et al. (2014): On Receiver Design for Diffusion-Based Molecular Communication

Paper 2

Bilge et al. (2025): Reservoir Computing-Based Detection for Molecular Communications

The Core Challenge: Inter-Symbol Interference (ISI)

How It Works

Concentration Shift Keying

(OOK): Bit-1 = release molecules;
Bit-0 = silence.

Molecules travel via **stochastic Brownian motion** — they don't disappear after detection. They linger, creating a **heavy tail** in the Channel Impulse Response (CIR).

The Result

Residual molecules from previous bits corrupt the next signal — **ISI**.

The Danger

Faster transmission → shorter symbol gaps → chemical echo overwhelms the receiver.



Paper 1 (Meng 2014): ISI Cancellation

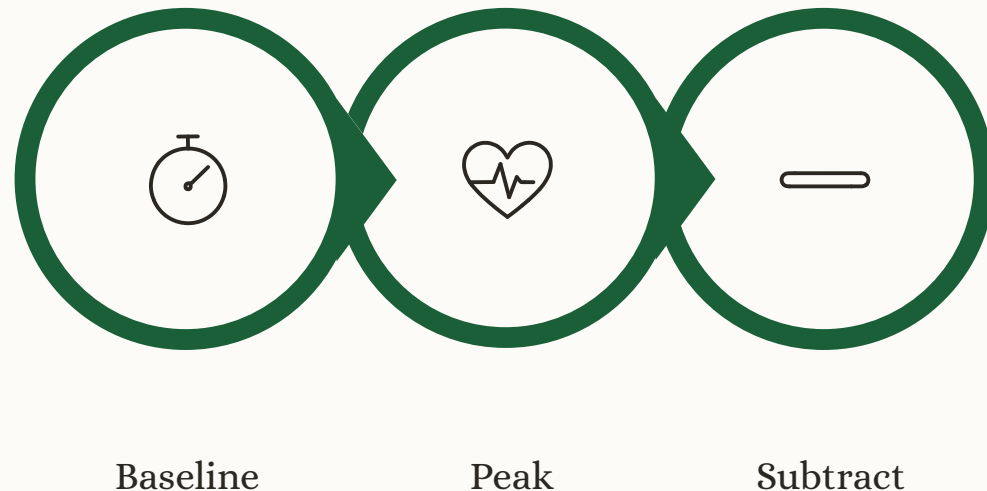
⚗ STATIC CHANNEL SOLUTION

Core Assumption

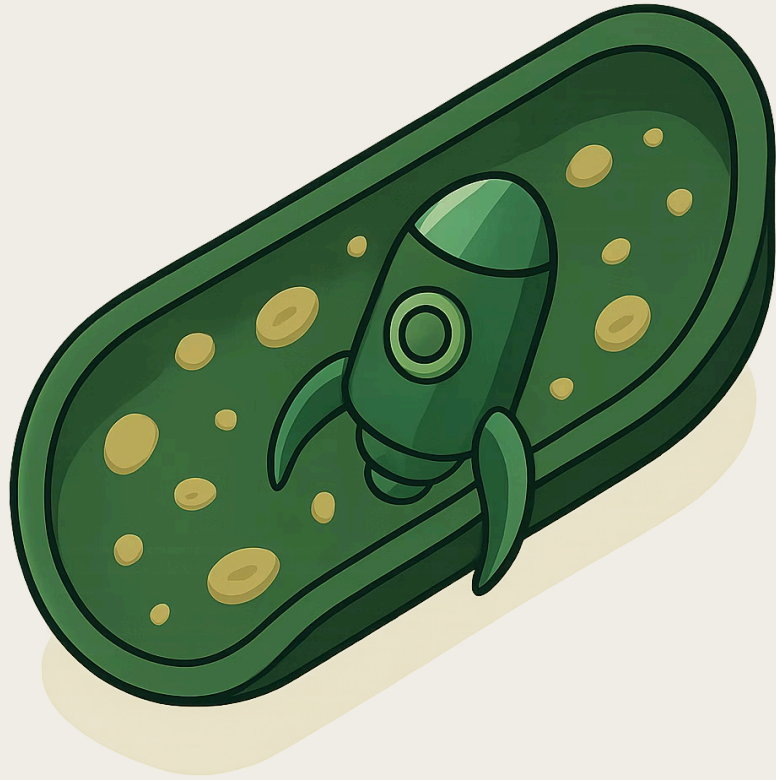
Static environment with ideal conditions
— no node mobility. Nano-devices have near-zero computational power, so heavy algorithms are off the table.

Why It's Elegant

A lightweight mathematical trick that requires only two concentration measurements and a single subtraction
— no training, no memory.



By isolating $\text{Signal} = \text{Peak} - \text{Baseline}$, the receiver cancels accumulated molecular history and recovers the transmitted bit without expensive computation.



The Mobile Channel Problem

Meng's solution is mathematically elegant — but assumes the nodes are **stationary**. Real biological environments break this assumption entirely.



Mobile Nano-Nodes

In blood vessels, nodes drift continuously due to Brownian motion, changing TX-RX distance every second.



Time-Varying CIR

A shifting distance means the channel impulse response is never the same — fixed mathematical thresholds collapse.



Need for Adaptation

We need a system that **learns the channel dynamically** — transitioning from physics-based math to a data-driven approach.

Paper 2 (Bilge 2025): Reservoir Computing

The ML Dilemma

Deep learning adapts well to dynamic channels — but **LSTMs and CNNs** are far too resource-heavy for nanoscale devices. Insufficient battery, memory, and compute rule them out.

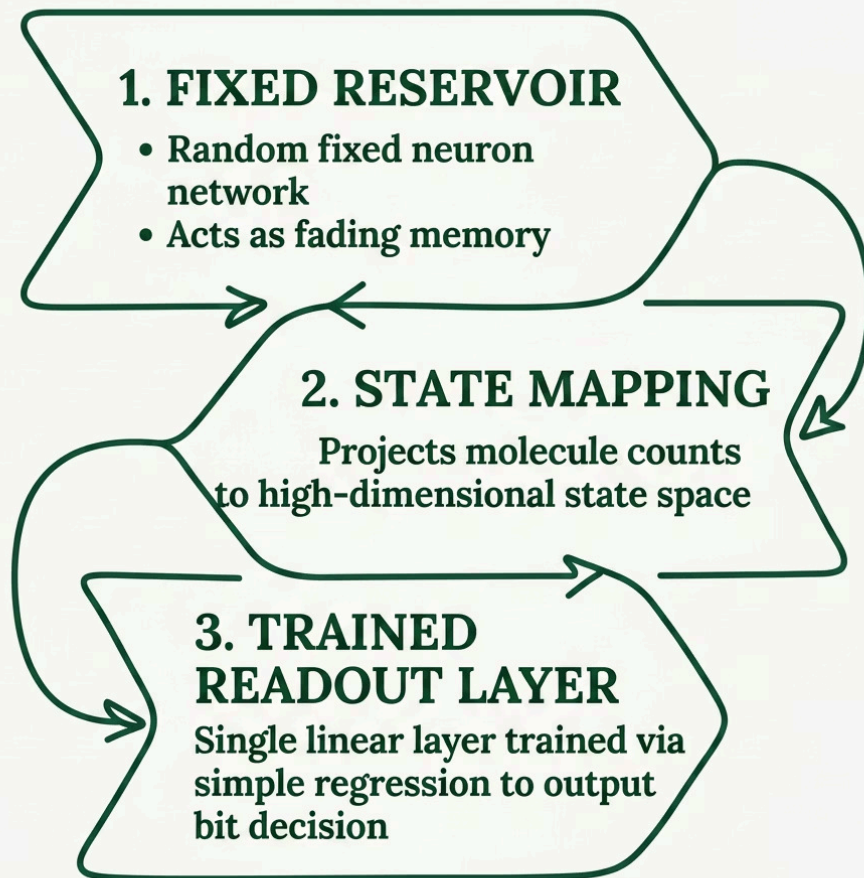
The Proposed Solution

Reservoir Computing (RC) retains the temporal memory of a recurrent neural network while drastically reducing training complexity — making it a viable candidate for nano-device deployment.

Why RC Wins

- Handles time-varying, non-stationary channels
- Requires training only on a single linear readout layer
- No backpropagation through time
- Ultra-low parameter count and inference latency

Reservoir Computing Architecture



The Pool Analogy

Think of the reservoir as a pool of water. Incoming molecules are like stones dropped in — their ripples mix with older ripples, naturally encoding temporal history.

- ✓ Only the final readout layer is trained — using simple linear regression. No backpropagation. Training cost is near zero.

Simulation Environment: Smoldyn

Framework

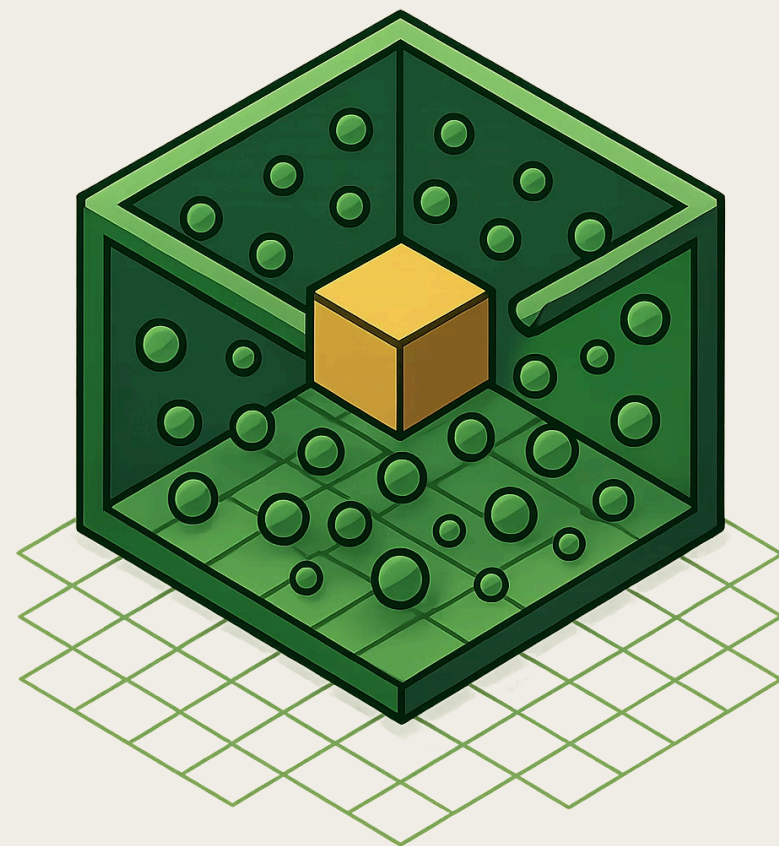
Smoldyn – 3D particle-based spatial stochastic simulator with mobile TX and RX nodes.

Transmission

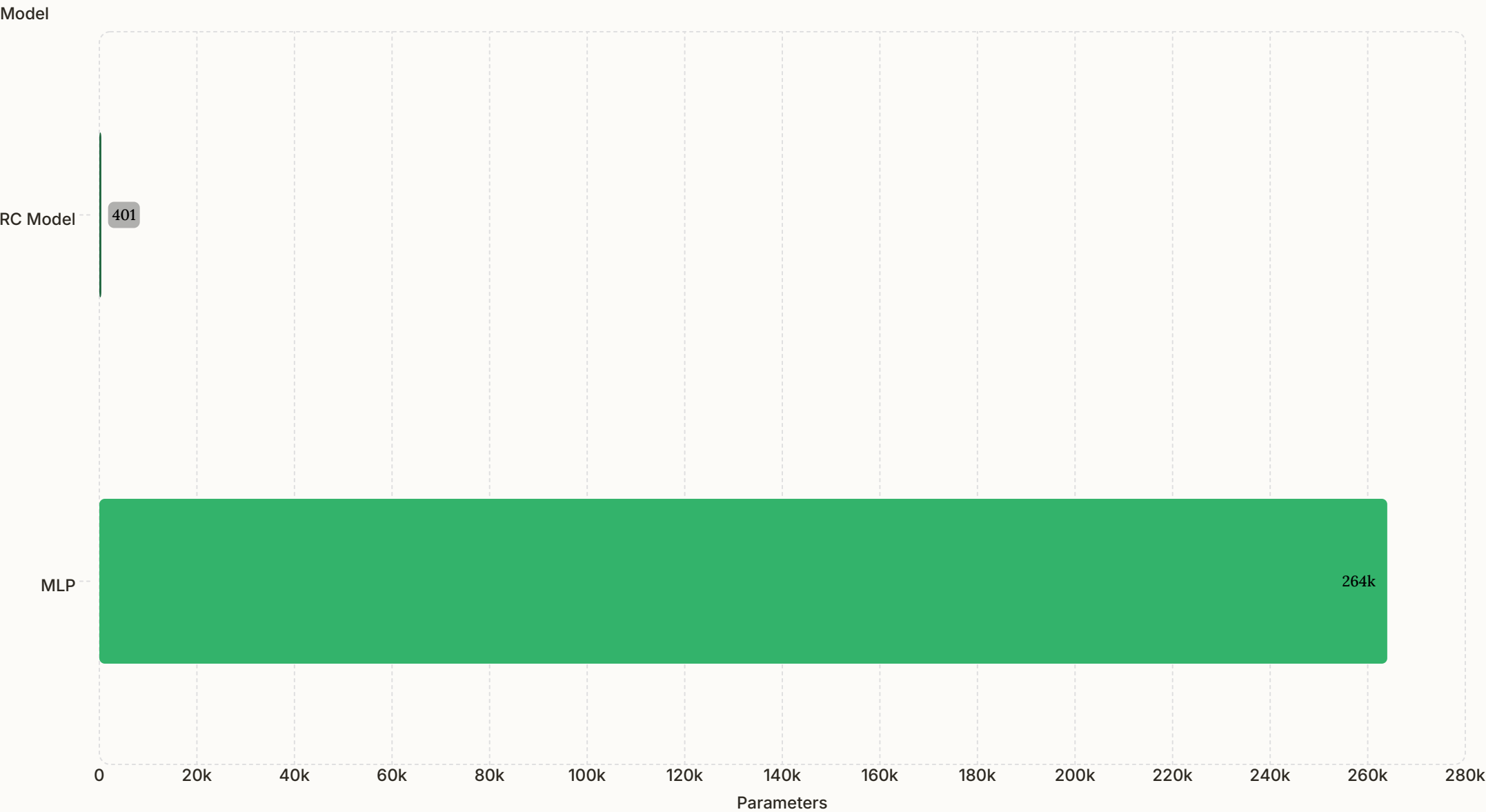
Bit-1 releases **2,000 molecules** per symbol interval into the bounded 3D environment.

ISI Stress Test

Symbol duration T_b varied from **200s** (low ISI) down to **10s** (extreme ISI).



Superior Results & Efficiency



At Extreme ISI ($T_b = 10s$)

LSTM and CNN models drop to **~50% accuracy** — effectively random guessing. The RC model maintains robustness where others fail.

401

RC Parameters
vs. MLP's 264,000

~1 μ s

Per-Symbol Latency
Ultra-fast inference

Our Final Project: Hybrid RC Extension

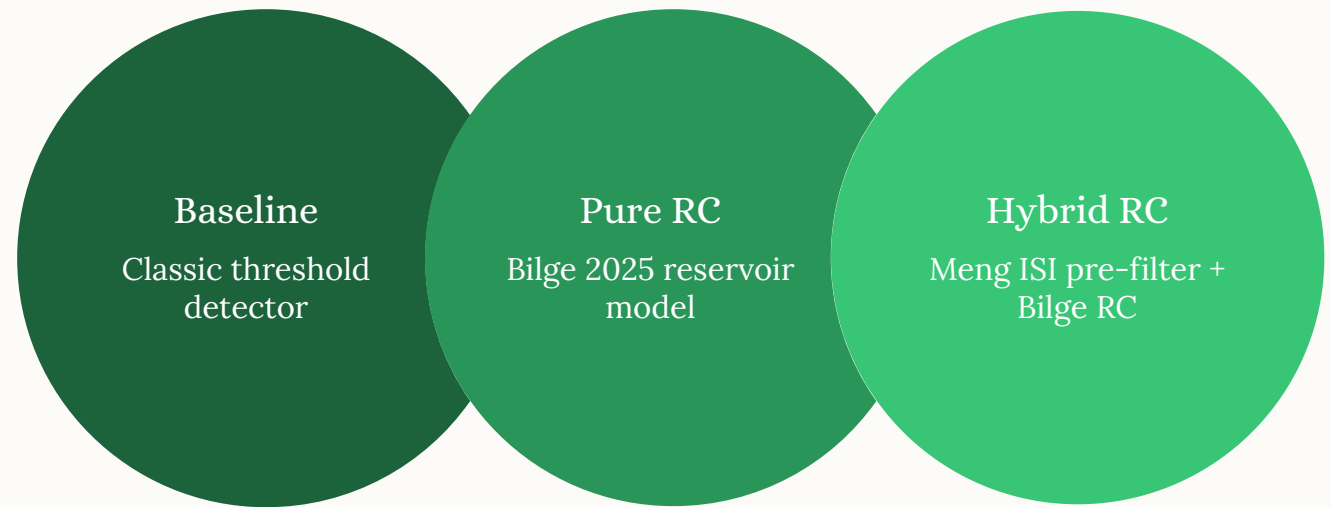
TASK 3 PROPOSAL

Objective

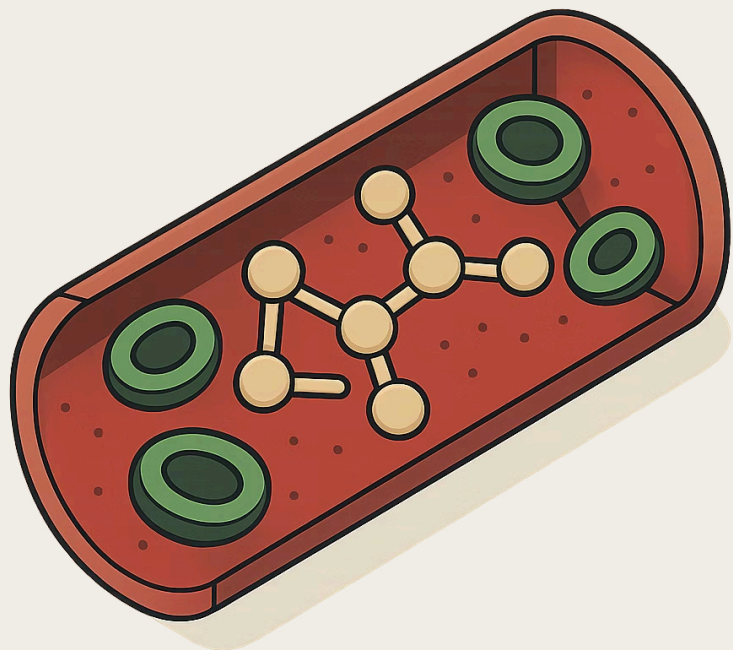
Build a Python-based particle simulator and implement the RC detection model from scratch – then introduce two novel extensions.

Extension 1 – Hybrid Pre-processing: Apply Meng's ISI Cancellation as a mathematical filter before feeding cleaned data into the RC model.

Extension 2 – Multi-Receiver Consensus: Simulate multiple nano-receivers combining their RC readouts for a joint bit decision.



We will benchmark all three configurations and present accuracy metrics as our final project.



Conclusion & Thank You

→ The Problem

Molecular communication suffers from severe ISI due to Brownian motion and slow molecular diffusion.

→ Classical Solution

Meng (2014) provides a lightweight mathematical fix — elegant, but limited to static channels.

→ ML Solution

Reservoir Computing (Bilge 2025) delivers dynamic adaptability with minimal hardware cost — only 401 parameters.

We're excited to combine physics-based math and modern ML in our own simulation. Questions and feedback are very welcome!